the ovary. The secretion of both male and female hormones are under the control of the pituitary gonadotropins luteinizing hormone (LH) and follicle stimulating hormone (FSH).

17.9 Menstrual Cycle-Process of Ovulation

The cyclic events that take place in a rhythmic fashion during the reproductive period of a woman's life is called menstrual cycle. In human females the menstrual cycle starts at the age of 11-13 years which marks the onset of puberty and is called **menarche**, and ceases around 48-50 years of age and this stage is termed **menopause**. The reproductive period is marked by characteristic events repeated almost every month in physiologically normal women (28 days with minor variation) in the form of a menstrual flow. The menstrual cycle consists of 4 phases

1. Menstrual or Destructive Phase
2. Follicular or Proliferative Phase
3. Ovulatory Phase
4. Luteal or Secretory Phase

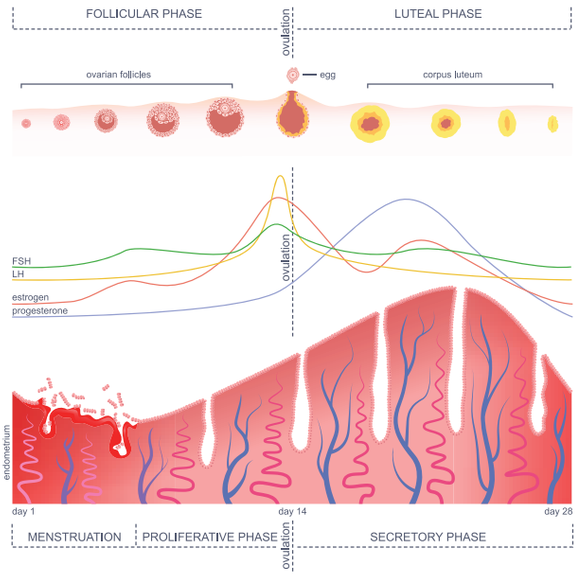


Figure 17.17 Menstrual cycle

These phases show simultaneous synchrony of events in both ovary and uterus. Changes in the ovary and the uterus are

Table 17.1 Events of Menstrual Cycle and the Role of Hormones

Phase	Days	Changes in Ovary	Changes in Uterus	Hormonal Changes
Menstrual phase	4–5 days	Development of primary follicles	Breakdown of uterine endometrial lining leads to bleeding	Decrease in progesterone and oestrogen
Follicular phase	6th–13th day	Primary follicles grow to become a fully mature Graafian follicle	Endometrium regenerates through proliferation	FSH and oestrogen increase
Ovulatory phase	14th day	The Graafian follicle ruptures, and releases the ovum(egg)	Increase in endometrial thickness	LH peak
Luteal phase	15th–28th day	Emptied Graafian follicle develops into corpus luteum	Endometrium is prepared for implantation if fertilization of egg takes place, if fertilization does not occur corpus luteum degenerates, uterine wall ruptures, bleeding starts and unfertilized egg is expelled	LH and FSH decrease, Corpus luteum produces progesterone and its level increases followed by a decline, if menstrual bleeding occurs

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